

GEN-AI :Unit1-Submission 2-Project1

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Section :D

problem statement :Customer support bot that answers questions based on a company's policy document.

Abstract:Q&A

- **How many days of paid time off do employees get?** The chatbot responded that "employees are eligible for 20 days of paid time off per year," with unused paid time off being carried over for a maximum of 5 days into the next year. It also included information about sick leave.
- **Can I carry over sick leave?** The chatbot stated that sick leave "cannot be carried over," as employees are allotted 10 days per year. It also mentioned that unused paid time off can be carried over for a maximum of 5 days and that company equipment must be returned upon termination of employment.
- **What is the policy on remote work and equipment return?** The chatbot indicated that "all company equipment must be returned upon termination of employment" and "remote work is permitted for up to two days a week, with prior manager approval." It also included a sentence about paid time off eligibility.
- **Is the chatbot functional?** Yes, the chatbot demonstrates basic functionality in retrieving relevant sentences from the policy document to answer direct questions.

Data Analysis Key Findings

- The chatbot successfully retrieved the correct number of paid time off days (20 days) and its carry-over policy, along with the non-carry-over policy for sick leave.
- For the sick leave carry-over question, the chatbot accurately stated that sick leave "cannot be carried over."
- The chatbot correctly identified policies for remote work (up to two days a week with manager approval) and equipment return (upon termination of employment).
- While generally accurate, the chatbot's responses sometimes included less directly relevant information, such as details about sick leave when asked about PTO, or PTO when asked about remote work/equipment return. This indicates some noise in the retrieval process.

Insights or Next Steps

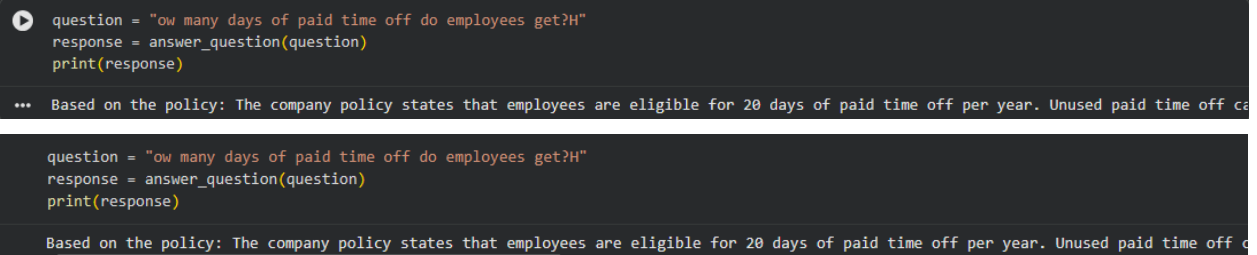
- Refine the retrieval mechanism or implement post-processing to minimize the inclusion of less pertinent information in the chatbot's answers, ensuring only the most highly relevant sentences are presented.
- For more complex queries requiring synthesis or inference beyond direct sentence retrieval, consider integrating a more advanced Question-Answering model on top of the current context retrieval system.

short documentation explaining what you understood and what you built:

This project implements an Automated FAQ Bot that answers user questions directly from a given policy document using Natural Language Processing techniques. The document is first split into smaller meaningful units, and semantic embeddings are generated to

capture contextual meaning. When a user asks a question, the system retrieves the most relevant content and applies an extractive Question Answering model to return accurate, document-grounded responses. The solution avoids hallucination by restricting answers strictly to the provided text and demonstrates the core concepts behind Retrieval-Augmented Generation (RAG). Built using Hugging Face Transformers, Sentence Transformers, and NLTK on Google Colab, the project is scalable and can be extended to support multiple documents, vector databases, and web-based interfaces.

Sample output/ screen shots :



```
question = "ow many days of paid time off do employees get?H"
response = answer_question(question)
print(response)
```

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